

NASSAU CANDY DISTRIBUTOR

Exploratory Data Analysis & Business Intelligence Report

Dataset Period: January 2024 – December 2025

Records Analyzed: 10,194 Transactions | 15 Products | 5,044 Customers

Prepared by: Business Analytics Division



1. Executive Summary

This report presents a comprehensive Exploratory Data Analysis (EDA) of Nassau Candy Distributor's transactional data spanning January 2024 through December 2025. The dataset comprises 10,194 order lines across 8,549 unique orders, covering 15 distinct products sold to 5,044 customers in 542 cities across the United States. The analysis uncovers sales trends, profitability drivers, regional dynamics, product performance, and customer behavior patterns to inform strategic decision-making.

\$141,783	\$93,443	66.5%	5,044	\$16.58
Total Revenue (2024–2025)	Total Gross Profit	Avg Gross Margin	Unique Customers	Avg Order Value

Key Highlights:

- Total revenue grew 44.6% year-over-year from \$57,956 (2024) to \$83,827 (2025).
- Chocolate division dominates with 92.8% of all revenue and commands the highest margin at 67.5%.
- Wonka Bars account for ~78.8% of all sales — making them the business-critical product line.
- Pacific region leads in both sales (\$46,302) and volume (3,253 orders), followed by Atlantic.
- Q4 seasonality is pronounced — November and December consistently show the highest monthly sales.
- Kazookles is a severe margin outlier at only 7.7% — immediate action required.
- "Other" division generates \$9,663 in revenue but with thin 37.7% margins needing review.

2. Dataset Overview

The Nassau Candy Distributor dataset is a transaction-level sales record containing 18 fields per row. The data is complete with zero missing values across all 10,194 records — a strong indicator of mature data collection infrastructure. The 24-month time window enables robust year-over-year analysis.

2.1 Data Schema

Field	Type	Description
Row ID	Integer	Unique row identifier
Order ID	String	Unique order reference (encodes year, customer, product)
Order Date	Date	Date order was placed (Jan 2024 – Dec 2025)
Ship Date	Date	Date order was shipped (extends to mid-2026 for late-2025 orders)
Ship Mode	Categorical	Standard Class, Second Class, First Class, Same Day
Customer ID	Integer	Unique customer identifier (range: 100,006–192,314)
Country/Region	String	All records: United States
City / State	String	542 cities across 59 state/territory entries
Division	Categorical	Chocolate (96.6%), Other (3.0%), Sugar (0.4%)
Region	Categorical	Pacific, Atlantic, Interior, Gulf
Product ID / Name	String	15 unique products across 3 divisions
Sales	Float	Revenue per line item (\$1.25 – \$260.00)
Units	Integer	Quantity sold per line item
Gross Profit	Float	Profit after cost of goods sold
Cost	Float	Cost of goods sold per line item

2.2 Data Quality Assessment

- Missing Values: 0 across all 18 columns — dataset is 100% complete.
- Date Coverage: Full 24-month window enabling strong temporal analysis.
- Ship Date Anomaly: Some ship dates extend into 2026 — logistically valid for late-2025 orders.
- Geographic Scope: 542 cities, 59 state entries — entirely United States.

3. Exploratory Data Analysis

3.1 Sales Trend Analysis

Monthly sales show a clear upward trajectory over 24 months. 2024 started modestly at \$2,094 in January and peaked at \$8,918 in November. In 2025, every month outperformed its 2024 counterpart. September, November, and December 2025 each exceeded \$11,000, with December 2025 setting the all-time monthly high at \$12,338. The pattern confirms strong Q4 seasonality driven by the holiday confectionery cycle.

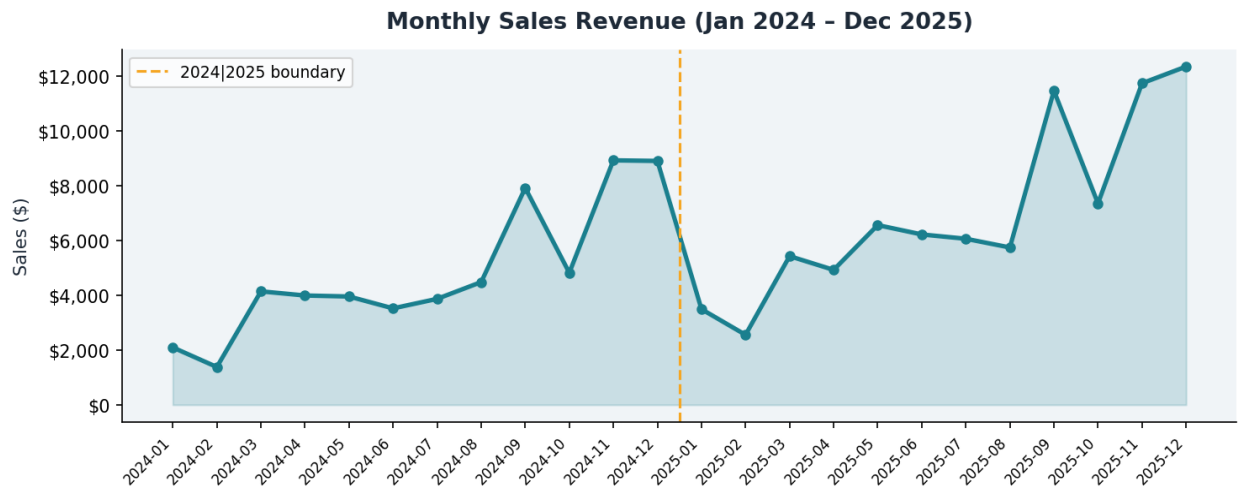


Fig 1. Monthly Sales Revenue — January 2024 to December 2025

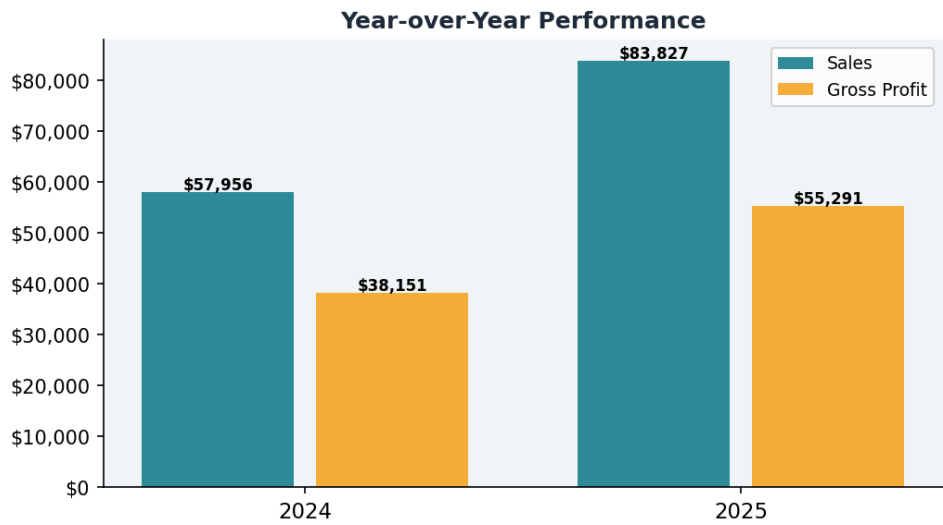


Fig 2. Year-over-Year Revenue and Gross Profit Comparison

Metric	2024	2025	Growth
Total Sales	\$57,956	\$83,827	+44.6%
Gross Profit	\$38,151	\$55,291	+44.9%
Units Sold	15,899	22,755	+43.1%
Gross Margin %	65.8%	66.0%	+0.2 pp

Insight: Revenue grew 44.6% while margins held steady — a sign of healthy, scalable growth rather than margin-dilutive discounting. Both units and revenue grew proportionally.

3.2 Division Performance

Nassau Candy operates across three product divisions. The concentration in Chocolate is extreme — 96.6% of all order lines and 92.8% of revenue. The Other division (specialty novelty items including Lickable Wallpaper) contributes meaningfully at \$9,663 but at a significantly lower margin.

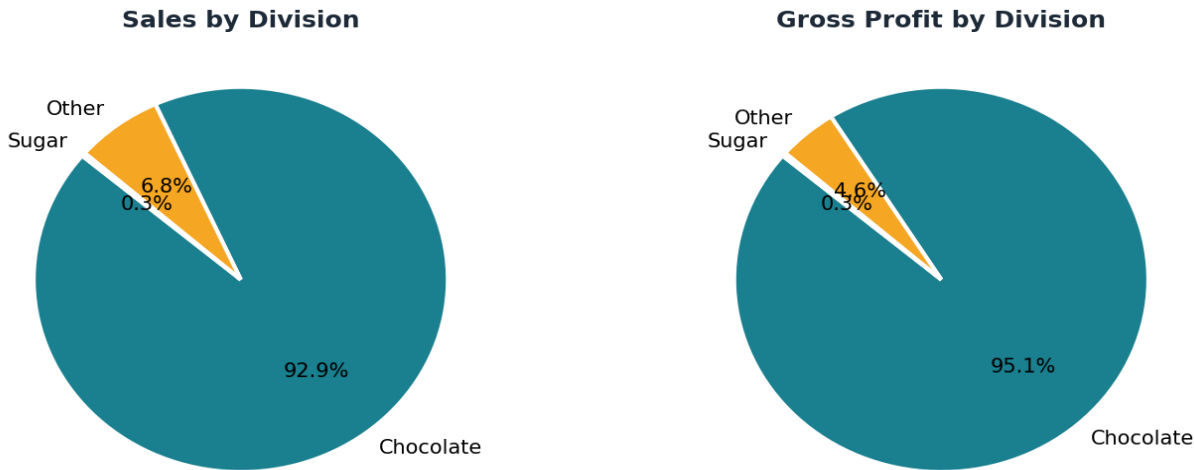


Fig 3. Revenue and Gross Profit Distribution by Division

Division	Sales	Gross Profit	Margin %	Units	Orders
Chocolate	\$131,693	\$88,825	67.5%	37,275	9,844
Other	\$9,663	\$4,333	37.7%	1,242	310
Sugar	\$427	\$285	57.7%	137	40

Insight: The Other division runs at 37.7% margin — 30 percentage points below Chocolate. Pricing review or cost renegotiation for this division is a priority.

3.3 Product Performance

The 15-SKU catalog is heavily concentrated in five Wonka Bar flavors, which collectively generate \$111,692 in revenue (78.8% of total). Product-level margin analysis reveals extreme variation — from 80% (Everlasting Gobstopper) to 7.7% (Kazookles).

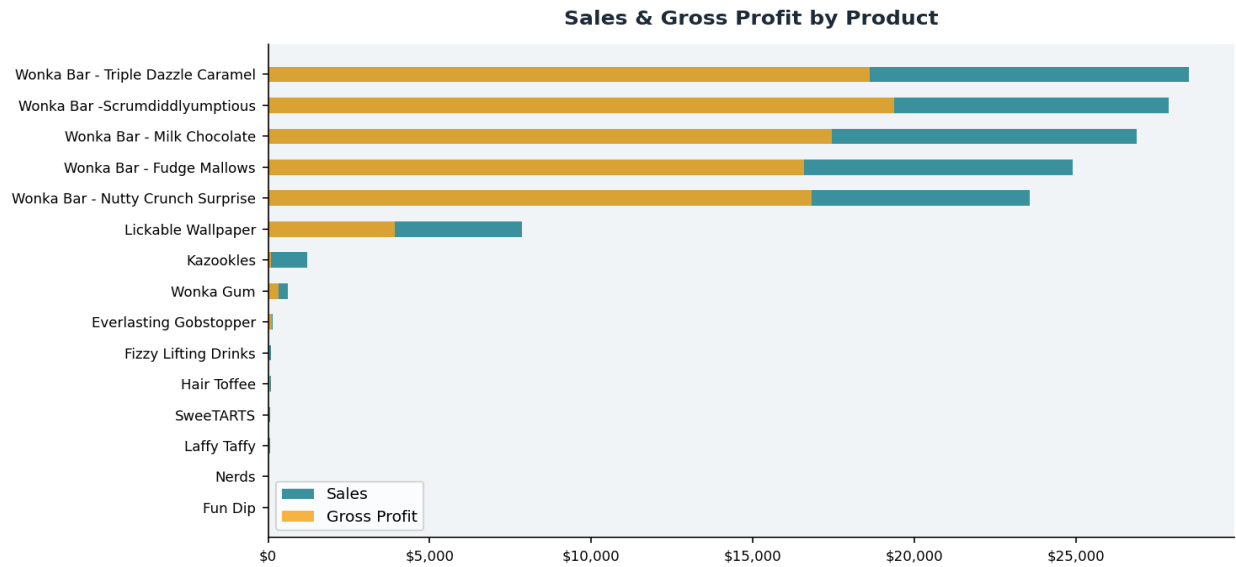


Fig 4. Total Sales and Gross Profit by Product

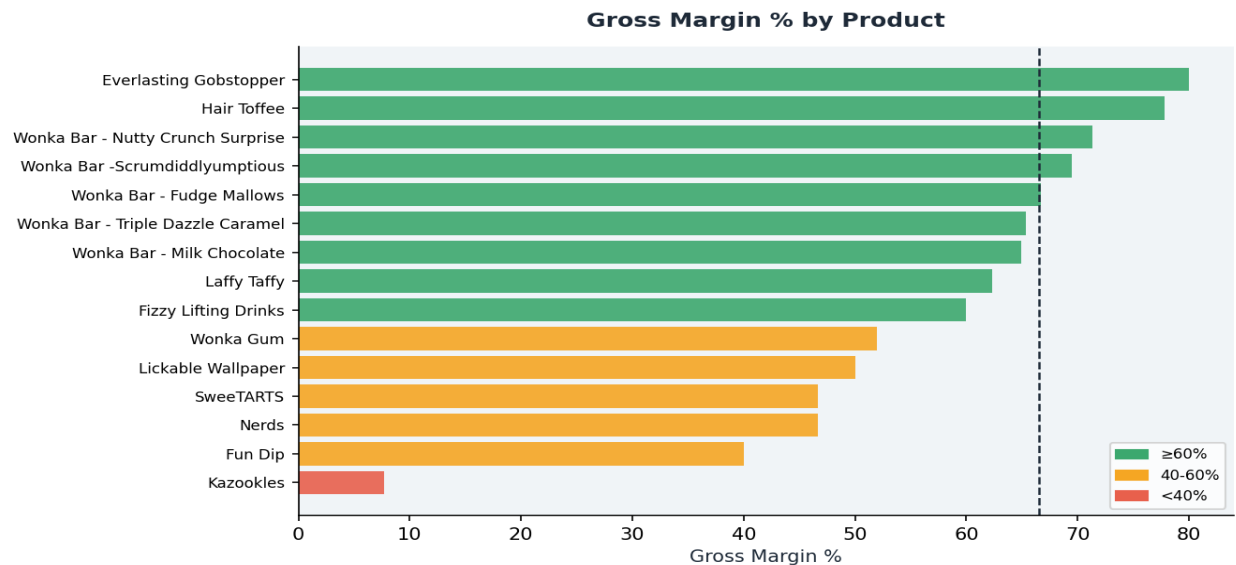


Fig 5. Gross Margin % by Product (green ≥ 60%, amber 40–60%, red < 40%)

Product	Sales	GP	Margin%	Units
Wonka Bar - Triple Dazzle Caramel	\$28,485	\$18,610	65.3%	7,596
Wonka Bar - Scrumdiddlyumptious	\$27,875	\$19,358	69.4%	7,743
Wonka Bar - Milk Chocolate	\$26,868	\$17,443	64.9%	8,267
Wonka Bar - Fudge Mallows	\$24,890	\$16,594	66.7%	6,914
Wonka Bar - Nutty Crunch Surprise	\$23,575	\$16,820	71.3%	6,755
Lickable Wallpaper	\$7,860	\$3,930	50.0%	393
Kazookles	\$1,206	\$93	7.7%	371
Wonka Gum	\$598	\$311	52.0%	478

Product	Sales	GP	Margin%	Units
Everlasting Gobstopper	\$130	\$104	80.0%	13
Fizzy Lifting Drinks	\$79	\$47	60.0%	21

Insight: Kazookles: 371 units sold but only \$93 gross profit (7.7% margin). Urgent repricing or discontinuation required.

Insight: Everlasting Gobstopper has the highest margin (80%) but only 13 units sold — severely underdistributed with strong upside potential.

3.4 Regional Analysis

Four geographic regions serve the entire U.S. market. Pacific leads in revenue and volume. All regions maintain near-identical margins (~66%), indicating consistent pricing discipline with no regional discounting anomalies. California alone contributes 19.7% of total revenue.

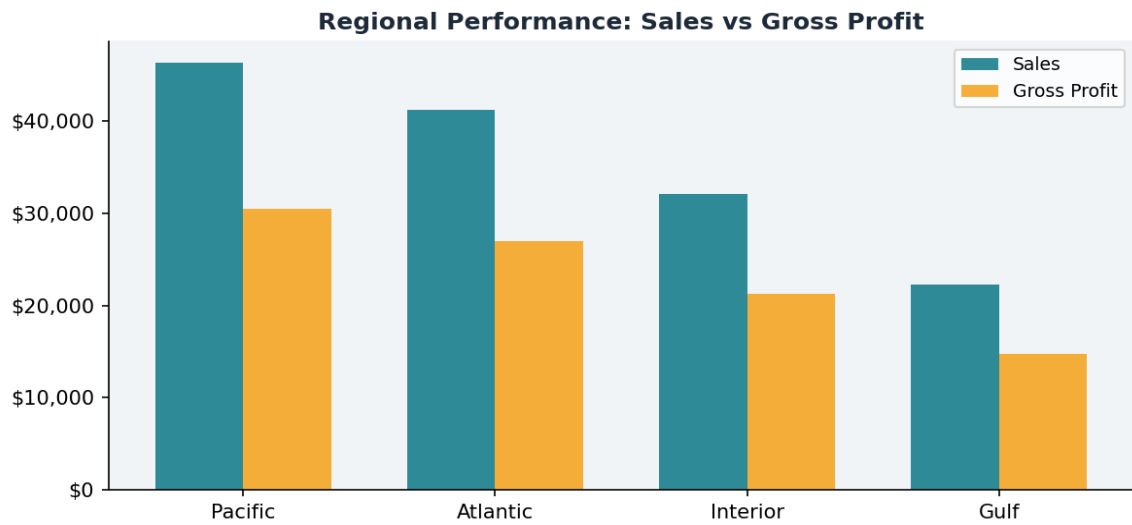


Fig 6. Regional Sales and Gross Profit Performance

Region	Sales	Gross Profit	Margin %	Units	Orders
Pacific	\$46,302	\$30,486	66.6%	12,466	3,253
Atlantic	\$41,197	\$26,974	66.2%	11,159	2,986
Interior	\$32,038	\$21,282	66.9%	8,820	2,335
Gulf	\$22,247	\$14,701	66.4%	6,209	1,620

Top 5 States by Revenue:

Rank	State	Sales	Gross Profit	Region
1	California	\$27,917	\$18,479	Pacific
2	New York	\$15,541	\$10,222	Atlantic
3	Texas	\$13,416	\$8,910	Gulf/Interior
4	Pennsylvania	\$8,027	\$5,225	Atlantic
5	Washington	\$6,921	\$4,567	Pacific

Insight: California drives 19.7% of total revenue — a geographic concentration risk. The Gulf region is significantly underpenetrated relative to its market size.

3.5 Shipping Mode and Order Analysis

Ship Mode	Orders	Sales	Gross Profit	Avg Margin
Standard Class	6,120	\$85,490	\$56,424	66.5%
Second Class	1,979	\$27,860	\$18,307	66.5%
First Class	1,548	\$21,319	\$14,011	66.5%
Same Day	547	\$7,114	\$4,701	66.5%

Order Metric	Value
Total Unique Orders	8,549
Average Order Value	\$16.58
Median Order Value	\$13.00
Avg Line Items per Order	1.19 items
Maximum Single Order	\$260.00

Insight: Average order contains only 1.19 line items — most orders are single-product. Multi-product bundling is an immediate, low-cost revenue growth lever.

4. Key Business Insights

Revenue Growing Strongly (+44.6% YoY)	A 44.6% revenue increase in a single year is exceptional. Growth is consistent across all regions and products, suggesting organic demand expansion. Trend shows acceleration in H2 of each year, likely amplified by holiday season purchasing.
Chocolate = The Entire Business	With 92.8% of revenue from Chocolate and ~78.8% from Wonka Bars alone, Nassau Candy is effectively a Wonka Bar distribution company. This focus enables expertise but creates severe concentration risk. Any supply or demand disruption to Wonka Bars would be existential.
Q4 Seasonality is Pronounced	October, November, and December consistently represent peak revenue months. December 2025 hit \$12,338 — nearly 6x January 2024. Holiday confectionery demand requires proactive inventory planning 3–4 months in advance to avoid stockouts.
Kazookles is a Profitability Crisis	Kazookles generated \$1,206 in revenue from 371 units but only \$92.75 in gross profit (7.7% margin). This product almost certainly does not cover its selling and distribution costs. It is the single most urgent profitability problem in the portfolio.
Gulf Region is Significantly Underpenetrated	The Gulf region contributes only 15.7% of revenue. New York state alone (\$15,541) nearly matches the entire Gulf region (\$22,247). Untapped customer potential in TX, FL, and LA represents a meaningful near-term growth opportunity.
Customer Base is Highly Fragmented	With 5,044 unique customers averaging only \$28.10 in total spend each, no single customer dominates. The top customer spent just \$454. This creates low customer concentration risk but makes account management expensive and customer LTV very low.

5. Strategic Recommendations

IMMEDIATE ACTION

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| 1. Discontinue or Reprice Kazookles | At 7.7% gross margin, Kazookles destroys value. Renegotiate supplier costs to achieve 40%+ margin, raise retail price, or discontinue. This is the single highest-impact quick win available. |
| 2. Audit Other Division Pricing | The Other division runs at 37.7% margin — 30 points below Chocolate. Novelty items like Lickable Wallpaper may support premium pricing. Conduct pricing analysis and implement 15–20% price increases where demand elasticity allows. |

SHORT-TERM (0–6 months)

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| 1. Expand Everlasting Gobstopper Distribution | At 80% gross margin with only 13 units sold, this product is severely underdistributed. Pilot expanded distribution in Pacific and Atlantic regions. Even 500 units would add \$5,200 in gross profit at current pricing. |
| 2. Launch Multi-Product Bundle Promotions | Average order contains only 1.19 line items. Introduce Wonka Bar variety packs and cross-division bundles. A 20% lift in average order value would add ~\$28,000 annually at current order volume. |
| 3. Gulf Region Growth Initiative | Commission a market study to assess Gulf region potential. Target 200+ new accounts in Texas, Florida, and Louisiana within 6 months. Consider adding a dedicated regional sales representative. |

MEDIUM-TERM (6–18 months)

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| 1. Diversify Product Portfolio | Wonka Bar concentration (78.8% of revenue) is existential risk. Actively recruit 2–3 new confectionery brands with gross margins of 55%+ and national brand recognition. Target 70% Wonka / 30% other within 18 months. |
| 2. Implement Q4 Demand Planning Process | With November–December running 2–3x January volumes, supply chain must be planned 3–4 months ahead. Establish pre-season inventory commitments with suppliers and build safety stock protocols. |
| 3. Customer Loyalty and Retention Program | With average customer lifetime value of only \$28.10, a structured reorder incentive program is essential. Target a 15% reduction in single-order customers. Introduce tiered volume discounts for accounts ordering 3+ times annually. |
| 4. Geographic Risk Mitigation | California drives 19.7% of revenue. Actively grow representation in Florida, Georgia, and the Midwest to reduce single-state concentration below 15%. Set state-level sales targets for underperforming markets. |

6. Conclusion

Nassau Candy Distributor is a growing, profitable confectionery distribution business with a clear dominant product (Wonka Bars), strong geographic footprint, and exceptional margin consistency. The 44.6% revenue growth from 2024 to 2025 demonstrates real market momentum and operational effectiveness.

However, the business faces three structural vulnerabilities: extreme product concentration (one brand family at 78.8% of revenue), significant geographic concentration (California at 19.7% of revenue), and a fragmented, low-LTV customer base averaging only \$28 in lifetime spend. Addressing these through portfolio diversification, Gulf region expansion, and customer loyalty investment should be the strategic priorities for the next 12–18 months.

At the product level, Kazookles requires urgent action, while Everlasting Gobstopper represents the most underexploited high-margin opportunity. The Other division has revenue potential but needs margin improvement to justify continued investment.

Overall, Nassau Candy is well-positioned for continued growth. The consistent ~66.5% gross margin across all regions and shipping modes signals disciplined pricing. With targeted interventions in product mix, geographic reach, and customer engagement, the business can sustain and accelerate its current growth trajectory into 2026 and beyond.

This report was generated through systematic analysis of 10,194 transactional records. All figures are derived from the Nassau Candy Distributor dataset (Jan 2024 – Dec 2025). Recommendations represent analytical conclusions and should be validated with operational stakeholders before implementation.